

ENERGY: sMAPE at each forecast step
Lookback = 72 steps, Forecast horizon = 72 steps

Forecast step	CNN	LSTM	dCeNN+ELM	dCeNN+ELM+ASP
1.000	57.146	73.085	73.259	51.451
2.000	58.782	68.743	73.298	51.530
3.000	60.408	66.732	73.295	51.605
4.000	61.725	65.985	73.325	51.623
5.000	63.149	65.885	73.491	51.763
6.000	64.261	66.068	73.755	52.019
7.000	65.117	66.390	74.085	52.340
8.000	65.453	66.717	74.398	52.649
9.000	66.271	67.083	74.544	52.794
10.000	66.703	67.520	74.581	52.834
11.000	66.969	67.849	74.585	52.825
12.000	67.208	68.116	74.616	52.874
13.000	67.522	68.296	74.650	52.806
14.000	67.933	68.521	74.709	52.742
15.000	68.211	68.688	74.739	52.981
16.000	68.389	68.876	74.755	53.006
17.000	68.611	69.073	74.736	52.984
18.000	69.002	69.253	74.755	53.004
19.000	68.799	69.421	74.758	53.008
20.000	69.065	69.541	74.746	52.999
21.000	69.318	69.625	74.776	53.019
22.000	69.486	69.651	74.888	53.155
23.000	69.212	69.670	75.115	53.391
24.000	69.212	69.794	75.417	53.681
25.000	69.435	69.983	75.548	53.775
26.000	69.830	70.214	75.400	53.675
27.000	70.210	70.482	75.112	53.417
28.000	70.074	70.666	74.850	53.146
29.000	69.890	70.782	74.826	53.097
30.000	70.164	70.847	74.964	53.226
31.000	70.183	70.976	75.210	53.464
32.000	70.111	71.028	75.380	53.631
33.000	70.485	71.112	75.327	53.577
34.000	70.573	71.125	75.208	53.460
35.000	70.552	71.037	75.180	53.425
36.000	70.421	70.807	75.200	53.474
37.000	70.085	70.664	75.253	53.403
38.000	70.120	70.609	75.319	53.304
39.000	70.079	70.562	75.352	53.595
40.000	70.163	70.551	75.355	53.607
41.000	70.329	70.558	75.299	53.548
42.000	69.980	70.542	75.277	53.525
43.000	70.006	70.489	75.245	53.496
44.000	70.380	70.444	75.217	53.469
45.000	70.254	70.345	75.263	53.506
46.000	70.215	70.266	75.365	53.632
47.000	70.123	70.202	75.603	53.880
48.000	70.237	70.243	75.880	54.146
49.000	70.128	70.462	75.969	54.206
50.000	70.334	70.583	75.854	54.131
51.000	70.649	70.655	75.647	53.954
52.000	70.871	70.764	75.471	53.757
53.000	70.717	70.828	75.518	53.795
54.000	70.420	70.862	75.702	53.965
55.000	70.628	70.984	75.928	54.182
56.000	70.930	71.017	76.042	54.293
57.000	71.060	71.053	75.902	54.152
58.000	71.058	71.088	75.637	53.888
59.000	71.036	71.033	75.424	53.683
60.000	70.703	70.914	75.344	53.618
61.000	70.769	70.849	75.388	53.608
62.000	70.839	70.836	75.492	53.598
63.000	70.952	70.811	75.620	53.869
64.000	70.932	70.797	75.709	53.959
65.000	71.053	70.818	75.761	54.009
66.000	71.100	70.812	75.857	54.105
67.000	70.940	70.768	75.928	54.177
68.000	71.009	70.760	75.894	54.147
69.000	71.282	70.705	75.817	54.068
70.000	71.452	70.659	75.790	54.059
71.000	71.344	70.777	75.952	54.232
72.000	71.254	71.032	76.288	54.550

• This table shows how prediction error changes from the first forecast step to the last forecast step.
• It is especially useful for explaining whether a model degrades smoothly or sharply as it predicts further into the future.
• Lower is better. This is a percentage-style error that is easier to compare across scales.